

Topics covered :

Artificial Intelligence and Autonomous Systems in Space and Defense:

- The use of AI in space navigation, exploration, and decision-making.
- Autonomous systems in modern defense: drones, UAVs, and robotic warfare.

Satellite Technologies for Communication and Surveillance:

- Advanced satellite based communication systems
- Surveillance and reconnaissance satellites for defense intelligence.

Space Robotics and Exploration Technologies:

- Innovations in robotics for space exploration (rovers, robotic arms)

Cybersecurity for Space and Defense Infrastructure:

- Protecting space assets from cyber threats.
- Cyber defense strategies for critical defense infrastructure.

Geospatial Intelligence and Remote Sensing in Defense:

- Applications of remote sensing and GIS in modern defense strategies.
- Role of geospatial intelligence in enhancing situational awareness in defense operations.

OVERVIEW

The Faculty Development Program (FDP) on Advanced Solutions for Space Exploration and Modern Defense Strategies aims to provide faculty members, researchers, and professionals with insights into cutting-edge technologies driving space exploration and modern defense. This FDP will focus on innovative solutions that are shaping the future of space missions and defense systems, covering key areas such as artificial intelligence, robotics, satellite technology, cyber defense, and advanced materials.

OBJECTIVES

The objective of this FDP is to provide specialized training on advanced space exploration technologies, including AI, space robotics, and autonomous systems. It will also examine modern defence strategies and the impact of next-generation technologies on national security. Participants will gain insights into the critical roles of satellite communication, remote sensing, and geospatial intelligence in both space exploration and defence. The program will introduce materials and systems designed for extreme environments, such as those found in space and on the battlefield. Additionally, it will address cybersecurity challenges and offer practical solutions for safeguarding space assets and defence infrastructure, with real-world applications and industry connections for faculty, postgraduate students, research scholars, and industry professionals.



**Dept of Computer science engineering
and Dept of CSE – cyber security
jointly organising**

Six day online FDP on

**ADVANCED SOLUTIONS FOR
SPACE EXPLORATION AND
MODERN DEFENSE SYSTEM.**

December 2-7, 2024

Indian Institute of Information Technology Kottayam,
Valavoor P.O, Pala, Kottayam- 686635, Kerala.
www.iitkottayam.ac.in

RESOURCE PERSONS

- Dr.Vijay Varadharajan
Global Innovation Chair
Professor and Chief Cyber
Strategist, The University of
Newcastle, Australia
- Dr. Anil Kumar
Scientist Indian Institute of
Remote Sensing
- Dr. Ramachandra Prasad, IIIT
Hyderabad
- Dr. G. Mallikarjuna Rao
DRDO, Hyderabad, India
- Prof.Rama Rao Nidamanuri
IISST, Trivandram
- Dr. Anupam Sharma, Director,
DSP
- Dr. DRM Samudraiah, Deputy
Director, SDA
- Mr. Pavan Attavane, Director,
Advitiya R& D Labs
- V S S Kiran
Founder and CEO, Crunchbase
- Dr. Raj Seita, Scientist, Punjab
Remote Sensing Centre
- Dr. Bakimchandra Oinam, NIT
Manipur
- Dr. K. Krishna Naik, IIITDM,
Kurnool
- Dr. Malay Pramanik, Asian
Institute of Technology

COORDINATORS

- Dr A.Balu
Assistant Professor Department of
Cyber Security
- Dr A. Ravi
Assistant Professor Department
of CSE

PROGRAM CONVENERS

- Dr. Panchami V
Head of the Department of
Cyber security
- Dr. Rubell Marion Lincy G
Head of the Department of
Computer Science Engineering - II

PATRONS

- Prof. (Dr.) Prasad Krishna,
Director i/c, IIIT Kottayam.
- Dr. M.Radhakrishnan,
Registrar, IIIT Kottayam.
- Prof. (Dr.) P. Mohanan, Prof. In-
Charge (Academics), IIIT
Kottayam.

REGISTRATION

(Seats will be assigned on a first-come, first-served basis)

Target Audience :

Faculty, PhD scholars, UG/PG students
Industry Person

Registration Starts : 17-11-2024

Registration Closes : 30-11-2024

For Registration use the following link

[https://atalacademy.
aicte-india.org/signup](https://atalacademy.aicte-india.org/signup)

